Probability Theory

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Chapter 1

Foundations of Probability

1.1 Sample Space and σ-algebras

1.1.1 Definition The set of all possible outcomes of a random experiment is called the **sample space**, denoted by Ω. Each element $\omega \in \Omega$ is called an **outcome** (or **sample point**).

1.1.2 Example

- Flip a coin once:

$$\Omega = \{H, T\}.$$

- Flip a coin twice:

$$\Omega = \{HH, HT, TH, TT\}.$$

1.1.3 Definition The collection $\mathcal{F}$ of subsets of Ω is a σ-algebra on Ω if

(a) $\varnothing \in \mathcal{F}$.

(b) If

$$A_1, A_2, \ldots, A_n, \ldots \in \mathcal{F},$$

then

$$\bigcup_n A_n \in \mathcal{F}.$$

(c) If $A \in \mathcal{F}$, then

$$A^c \in \mathcal{F}.$$

An **event** is a subset of the sample space Ω.
An event A is called **measurable** if $A \in \mathcal{F}$.
$(\Omega, \mathcal{F})$ is called **measure space**.

1.1.4 Example Let

$$\Omega = \{HH, HT, TH, TT\}.$$

Consider the following collections of subsets of Ω:

(1)
$$\mathcal{F} = \{\varnothing, \Omega\}.$$

(2)
$$\mathcal{F} = 2^{\Omega}.$$

(3)
$$\mathcal{F} = \big\{\varnothing, \Omega, \{HH\}, \{HT, TH, TT\}\big\}.$$

(4)
$$\mathcal{F} = \big\{\varnothing, \Omega, \{HH, HT\}, \{TH, TT\}\big\}.$$

The σ-algebra specifies what information is available.

1.1.5 Proposition If $\mathcal{F}$ is a σ-algebra and $A_1, A_2, \ldots \in \mathcal{F}$, then

$$\bigcap_{n=1}^{\infty} A_n \in \mathcal{F}.$$

Proof By De Morgan's law,

$$\bigcap_{n=1}^{\infty} A_n = \left(\bigcup_{n=1}^{\infty} A_n^c\right)^c.$$

Since $A_n \in \mathcal{F}$, we have $A_n^c \in \mathcal{F}$ for every n. Hence

$$\bigcup_{n=1}^{\infty} A_n^c \in \mathcal{F},$$

and therefore

$$\left(\bigcup_{n=1}^{\infty} A_n^c\right)^c \in \mathcal{F}.$$

□

1.1.6 Exercise Let $\mathcal{F}$ be a σ-algebra on a finite sample space Ω. How many elements can $\mathcal{F}$ contain?

Solution Let $|\Omega| = n$. Since Ω is finite, every σ-algebra $\mathcal{F} \subseteq 2^\Omega$ is finite.
Let

$$B_1, \ldots, B_k$$

be the minimal nonempty elements of $\mathcal{F}$. These sets are pairwise disjoint and form a partition of Ω.
Moreover, every $A \in \mathcal{F}$ is a union of some of the B_i's. Conversely, every union of the B_i's belongs to $\mathcal{F}$.
Hence

$$\mathcal{F} = \left\{ \bigcup_{i \in I} B_i : I \subseteq \{1, \ldots, k\} \right\}.$$

There are 2^k subsets $I \subseteq \{1, \ldots, k\}$, so

$$|\mathcal{F}| = 2^k, \qquad 1 \leq k \leq n.$$

Therefore, the number of elements of a finite σ-algebra must be a power of 2.

1.1.7 Definition Given a measurable space $(\Omega, \mathcal{F})$, a **probability measure** on $(\Omega, \mathcal{F})$ is a function

$$\mathbb{P} : \mathcal{F} \longrightarrow [0, 1]$$

satisfying:

(a)

$$\mathbb{P}(\Omega) = 1.$$

(b) If $A_1, A_2, \ldots \in \mathcal{F}$ are pairwise disjoint, then

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n \right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n).$$

$(\Omega, \mathcal{F}, \mathbb{P})$ is called **probability space**.

1.1.8 Proposition Let $\mathbb{P}$ be a probability measure on $(\Omega, \mathcal{F})$. Then:

(1) $\mathbb{P}(\varnothing) = 0$.

(2) For every $A \in \mathcal{F}$,

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A).$$

(3) For every $A, B \in \mathcal{F}$,

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

Proof (1) Since $\varnothing$ and Ω are disjoint,

$$\mathbb{P}(\varnothing \cup \Omega) = \mathbb{P}(\varnothing) + \mathbb{P}(\Omega) = \mathbb{P}(\Omega).$$

, hence

$$\mathbb{P}(\varnothing) = 0.$$

(2) Since $A \cap A^c = \varnothing$ and $A \cup A^c = \Omega$,

$$1 = \mathbb{P}(\Omega) = \mathbb{P}(A \cup A^c) = \mathbb{P}(A) + \mathbb{P}(A^c),$$

so

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A).$$

(3) Write

$$A \cup B = A \cup (B \setminus A),$$

where A and $B \setminus A$ are disjoint. Hence

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A).$$

Also,

$$B = (B \setminus A) \cup (A \cap B)$$

is a disjoint union, so

$$\mathbb{P}(B) = \mathbb{P}(B \setminus A) + \mathbb{P}(A \cap B).$$

Therefore

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

□

1.1.9 Proposition Let $\mathbb{P}$ be a probability measure on $(\Omega, \mathcal{F})$.

(i) If

$$A_1 \subseteq A_2 \subseteq \cdots \subseteq A_n \subseteq \cdots,$$

then the sequence $\{\mathbb{P}(A_n)\}_{n\geq 1}$ is non-decreasing and

$$\lim_{n\to\infty} \mathbb{P}(A_n) = \mathbb{P}\left(\bigcup_{k=1}^{\infty} A_k\right).$$

(ii) If

$$B_1 \supseteq B_2 \supseteq \cdots \supseteq B_n \supseteq \cdots,$$

then the sequence $\{\mathbb{P}(B_n)\}_{n\geq 1}$ is non-increasing and

$$\lim_{n\to\infty} \mathbb{P}(B_n) = \mathbb{P}\left(\bigcap_{k=1}^{\infty} B_k\right).$$

Proof (i) Let

$$A = \bigcup_{k=1}^{\infty} A_k$$

and set $A_0 = \varnothing$. Since $A_1 \subseteq A_2 \subseteq \cdots$,

$$A = A_1 \cup (A_2 \setminus A_1) \cup (A_3 \setminus A_2) \cup \cdots,$$

where the sets on the right-hand side are pairwise disjoint. Hence

$$\mathbb{P}(A) = \sum_{k=1}^{\infty} \mathbb{P}(A_k \setminus A_{k-1}).$$

Therefore

$$\begin{aligned}\mathbb{P}(A) &= \lim_{n\to\infty} \sum_{k=1}^{n} \mathbb{P}(A_k \setminus A_{k-1}) \\ &= \lim_{n\to\infty} \mathbb{P}(A_n).\end{aligned}$$

Thus

$$\mathbb{P}(A_n) \uparrow \mathbb{P}\left(\bigcup_{k=1}^{\infty} A_k\right).$$

(ii) Since

$$B_1^c \subseteq B_2^c \subseteq \cdots,$$

part (i) gives

$$\mathbb{P}(B_n^c) \uparrow \mathbb{P}\left(\bigcup_{k=1}^{\infty} B_k^c\right).$$

Using De Morgan's law and $\mathbb{P}(B_n) = 1 - \mathbb{P}(B_n^c)$,

$$\begin{aligned}\lim_{n\to\infty} \mathbb{P}(B_n) &= 1 - \mathbb{P}\left(\bigcup_{k=1}^{\infty} B_k^c\right) \\ &= \mathbb{P}\left(\bigcap_{k=1}^{\infty} B_k\right).\end{aligned}$$

Hence

$$\mathbb{P}(B_n) \downarrow \mathbb{P}\left(\bigcap_{k=1}^{\infty} B_k\right).$$

□

1.1.10 Definition For a real sequence $(a_n)_{n\geq 1}$, define

$$\limsup_{n\to\infty} a_n := \inf_{n\geq 1} \sup_{k\geq n} a_k, \qquad \liminf_{n\to\infty} a_n := \sup_{n\geq 1} \inf_{k\geq n} a_k.$$

Equivalently, they are the limits of the decreasing tail suprema $\sup_{k\geq n} a_k$ and the increasing tail infima $\inf_{k\geq n} a_k$, respectively.

1.1.11 Definition For a sequence of sets $(A_n)_{n\geq 1}$, define

$$\limsup_{n\to\infty} A_n := \bigcap_{n=1}^{\infty} \bigcup_{k\geq n} A_k, \qquad \liminf_{n\to\infty} A_n := \bigcup_{n=1}^{\infty} \bigcap_{k\geq n} A_k.$$

Equivalently,

$$x \in \limsup_{n\to\infty} A_n \iff x \in A_n \text{ for infinitely many } n,$$

while

$$x \in \liminf_{n\to\infty} A_n \iff x \in A_n \text{ for all but finitely many } n.$$

In particular,

$$\liminf_{n\to\infty} A_n \subseteq \limsup_{n\to\infty} A_n.$$

1.1.12 Definition Let $A, B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$. The **conditional probability** of A given B is defined by

$$\mathbb{P}(A \mid B) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

1.1.13 Proposition Let $B_1, B_2, \ldots$ be pairwise disjoint events such that

$$\bigcup_k B_k = \Omega$$

and

$$\mathbb{P}(B_k) > 0 \qquad \text{for every } k.$$

Then, for every event A,

$$\mathbb{P}(A) = \sum_k \mathbb{P}(A \mid B_k)\mathbb{P}(B_k).$$

Proof Since the sets B_k are pairwise disjoint and $\bigcup_k B_k = \Omega$,

$$A = A \cap \Omega = A \cap \left(\bigcup_k B_k\right) = \bigcup_k (A \cap B_k),$$

where the sets $A \cap B_k$ are pairwise disjoint. Hence

$$\mathbb{P}(A) = \sum_k \mathbb{P}(A \cap B_k).$$

By the definition of conditional probability,

$$\mathbb{P}(A \cap B_k) = \mathbb{P}(A \mid B_k)\mathbb{P}(B_k).$$

Therefore

$$\mathbb{P}(A) = \sum_k \mathbb{P}(A \mid B_k)\mathbb{P}(B_k).$$

□

1.1.14 Definition Two events A and B are said to be **independent** if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

More generally, a sequence of events

$$A_1, A_2, \ldots$$

is said to be **independent** if, for every finite collection of distinct indices $k_1, \ldots, k_m$,

$$\mathbb{P}\left(\bigcap_{j=1}^m A_{k_j}\right) = \prod_{j=1}^m \mathbb{P}(A_{k_j}).$$

1.1.15 Remark For events $(A_k)_{k\geq 1}$,

$$\limsup_{k\to\infty} A_k$$

is the event that infinitely many of the A_k's occur.

1.1.16 Proposition Let $(A_k)_{k\geq 1}$ be a sequence of events.

(1) If

$$\sum_{k=1}^{\infty} \mathbb{P}(A_k) < \infty,$$

then

$$\mathbb{P}\left(\limsup_{k\to\infty} A_k\right) = 0.$$

(2) If

$$\sum_{k=1}^{\infty} \mathbb{P}(A_k) = \infty$$

and the events $A_1, A_2, \ldots$ are independent, then

$$\mathbb{P}\left(\limsup_{k\to\infty} A_k\right) = 1.$$

Proof (1) For every n,

$$\limsup_{k\to\infty} A_k \subseteq \bigcup_{k\geq n} A_k.$$

Hence

$$0 \leq \mathbb{P}\left(\limsup_{k\to\infty} A_k\right) \leq \mathbb{P}\left(\bigcup_{k\geq n} A_k\right) \leq \sum_{k=n}^{\infty} \mathbb{P}(A_k).$$

Letting $n \to \infty$, the tail sum tends to 0, so

$$\mathbb{P}\left(\limsup_{k\to\infty} A_k\right) = 0.$$

(2) For $m \geq n$, by independence,

$$\mathbb{P}\left(\bigcap_{k=n}^{m} A_k^c\right) = \prod_{k=n}^{m} \left(1 - \mathbb{P}(A_k)\right) \leq \exp\left(-\sum_{k=n}^{m} \mathbb{P}(A_k)\right).$$

Letting $m \to \infty$ gives

$$\mathbb{P}\left(\bigcap_{k=n}^{\infty} A_k^c\right) = 0.$$

Therefore

$$\mathbb{P}\left(\left(\limsup_{k\to\infty} A_k\right)^c\right) = \mathbb{P}\left(\bigcup_{n=1}^{\infty}\bigcap_{k=n}^{\infty} A_k^c\right) \leq \sum_{n=1}^{\infty} \mathbb{P}\left(\bigcap_{k=n}^{\infty} A_k^c\right) = 0.$$

Hence

$$\mathbb{P}\left(\limsup_{k\to\infty} A_k\right) = 1.$$

□

1.1.17 Example (Gambler's ruin) Consider a random walk on the states

$$\{0, 1, \ldots, N\}.$$

Starting from a state $k \in \{1, \ldots, N-1\}$, the walk moves

$$k \to k+1 \quad \text{with probability } p,$$

and

$$k \to k-1 \quad \text{with probability } q,$$

where

$$p + q = 1.$$

Let

$$\rho_k := \mathbb{P}\big(\text{hit } N \text{ before } 0 \,\big|\, \text{start at } k\big).$$

Then

$$\rho_0 = 0, \qquad \rho_N = 1.$$

Conditioning on the first step gives, for $1 \leq k \leq N-1$,

$$\rho_k = p\rho_{k+1} + q\rho_{k-1}.$$

Hence

$$p(\rho_{k+1} - \rho_k) = q(\rho_k - \rho_{k-1}),$$

so

$$\rho_{k+1} - \rho_k = \frac{q}{p}(\rho_k - \rho_{k-1}).$$

Therefore,

$$\rho_k - \rho_{k-1} = \left(\frac{q}{p}\right)^{k-1} (\rho_1 - \rho_0).$$

Since

$$1 = \rho_N - \rho_0 = \sum_{k=1}^{N} (\rho_k - \rho_{k-1}),$$

we obtain

$$1 = \left(1 + \frac{q}{p} + \left(\frac{q}{p}\right)^2 + \cdots + \left(\frac{q}{p}\right)^{N-1}\right) (\rho_1 - \rho_0).$$

(1) If

$$p = q = \frac{1}{2},$$

then

$$\rho_1 - \rho_0 = \frac{1}{N},$$

and therefore

$$\rho_k = \sum_{j=1}^{k} (\rho_j - \rho_{j-1}) = \frac{k}{N}.$$

(2) If $p \neq q$, then

$$\rho_1 - \rho_0 = \frac{1 - \frac{q}{p}}{1 - \left(\frac{q}{p}\right)^N}.$$

Hence

$$\rho_k - \rho_{k-1} = \left(\frac{q}{p}\right)^{k-1} \frac{1 - \frac{q}{p}}{1 - \left(\frac{q}{p}\right)^N},$$

and summing from 1 to k yields

$$\rho_k = \frac{1 - \left(\frac{q}{p}\right)^k}{1 - \left(\frac{q}{p}\right)^N}.$$

Thus

$$\boxed{\rho_k = \begin{cases} \dfrac{k}{N}, & p = q = \dfrac{1}{2}, \\ \dfrac{1 - \left(\frac{q}{p}\right)^k}{1 - \left(\frac{q}{p}\right)^N}, & p \neq q. \end{cases}}$$

1.1.18 Definition A **random variable** is a function

$$X : \Omega \to \mathbb{R}$$

such that, for every open set $A \subseteq \mathbb{R}$,

$$X^{-1}(A) = \{\omega \in \Omega : X(\omega) \in A\} \in \mathcal{F}.$$

Equivalently, X is $\mathcal{F}$-measurable.

1.1.19 Example Let

$$\Omega = \{HH, HT, TH, TT\}.$$

Define $X_1, X_2 : \Omega \to \mathbb{R}$ by

$$X_1(HH) = X_1(HT) = 1, \qquad X_1(TH) = X_1(TT) = -1,$$

and

$$X_2(HH) = X_2(TH) = 1, \qquad X_2(HT) = X_2(TT) = -1.$$

Let

$$X = X_1 + X_2.$$

Then the smallest σ-algebras that make X_1, X_2, and X random variables are, respectively,

$$\mathcal{F}_1 = \sigma\big(\{HH, HT\}, \{TH, TT\}\big),$$

$$\mathcal{F}_2 = \sigma\big(\{HH, TH\}, \{HT, TT\}\big),$$

and

$$\mathcal{F}_3 = \sigma\big(\{HH\}, \{HT, TH\}, \{TT\}\big).$$

1.1.20 Example Let

$$\Omega = \{1, 2, 3, 4, 5, 6\}, \qquad \mathcal{F} = \sigma\big(\{1, 3, 5\}, \{2, 4, 6\}\big).$$

The function

$$X_1(k) = -k$$

is not $\mathcal{F}$-measurable, while

$$X_2(k) = \begin{cases} 1, & k \text{ is odd}, \\ 2, & k \text{ is even}, \end{cases}$$

is $\mathcal{F}$-measurable.

1.1.21 Definition 1. Let $p \in [0,1]$. A random variable X has the **Bernoulli distribution** with parameter p, written

$$X \sim \mathrm{Ber}(p),$$

if

$$\mathbb{P}(X=1) = p, \qquad \mathbb{P}(X=0) = 1-p.$$

2. Let $n \in \mathbb{N}_+$ and $p \in [0,1]$. A random variable X has the **binomial distribution** with parameters (n,p), written

$$X \sim \mathrm{Bin}(n,p),$$

if

$$\mathbb{P}(X=k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, \ldots, n.$$

1.1.22 Remark If p is small and n is large with

$$np = \lambda > 0,$$

then, for fixed k,

$$\begin{aligned}\mathbb{P}(X=k) &= \binom{n}{k} \left(\frac{\lambda}{n}\right)^k \left(1-\frac{\lambda}{n}\right)^{n-k} \\ &\longrightarrow \frac{\lambda^k}{k!} e^{-\lambda}.\end{aligned}$$

1.1.23 Definition 1. Let $\lambda > 0$. A random variable X has the **Poisson distribution** with parameter λ, written

$$X \sim \mathrm{Pois}(\lambda),$$

if

$$\mathbb{P}(X=k) = e^{-\lambda} \frac{\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots.$$

2. Let $p \in (0,1]$. A random variable X has the **geometric distribution** with parameter p, written

$$X \sim \mathrm{Geo}(p),$$

if

$$\mathbb{P}(X = k) = (1-p)^{k-1}p, \qquad k \geq 1.$$

1.1.24 Proposition If

$$X \sim \text{Geo}(p),$$

then

$$\mathbb{P}(X > n) = (1-p)^n, \qquad n \geq 0.$$

Proof

$$\begin{aligned}\mathbb{P}(X > n) &= \sum_{k=n+1}^{\infty} p(1-p)^{k-1} \\ &= p(1-p)^n \sum_{k=0}^{\infty} (1-p)^k \\ &= (1-p)^n.\end{aligned}$$

□

1.1.25 Proposition A geometric random variable has the **memoryless property**:

$$\mathbb{P}(X > m+n \mid X > m) = \mathbb{P}(X > n).$$

Proof

$$\begin{aligned}\mathbb{P}(X > m+n \mid X > m) &= \frac{\mathbb{P}(X > m+n)}{\mathbb{P}(X > m)} \\ &= \frac{(1-p)^{m+n}}{(1-p)^m} \\ &= (1-p)^n \\ &= \mathbb{P}(X > n).\end{aligned}$$

□

1.1.26 Exercise Let X be a discrete random variable satisfying

$$\mathbb{P}(X > m+n \mid X > m) = \mathbb{P}(X > n)$$

for all $m, n \geq 0$. Show that

$$X \sim \text{Geo}(p)$$

for some $p \in (0, 1]$.

1.1.27 Remark (Stirling's formula)

$$n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n.$$

That is,

$$\lim_{n\to\infty} \frac{n!}{\sqrt{2\pi n}\left(\frac{n}{e}\right)^n} = 1.$$

Taking logarithms in Stirling's formula gives

$$\log(n!) = \frac{1}{2}\log(2\pi n) + n(\log n - 1) + o(1).$$

Hence,

$$\lim_{n\to\infty} \frac{\log(n!)}{n\log n} = 1.$$

1.1.28 Definition Let $X : \Omega \to \mathbb{R}$ be a random variable. The **distribution function** of X is

$$F : \mathbb{R} \to [0,1],$$

defined by

$$F(x) = \mathbb{P}(X \leq x).$$

If F is differentiable, then

$$f(x) := F'(x)$$

is called the **probability density function** of X.

1.1.29 Proposition The distribution function F is right continuous.

Proof Let

$$x_n \downarrow x.$$

Then

$$\{X \leq x_1\} \supseteq \{X \leq x_2\} \supseteq \cdots$$

and hence

$$\begin{aligned}\lim_{n\to\infty} F(x_n) &= \lim_{n\to\infty} \mathbb{P}(X \le x_n) \\ &= \mathbb{P}\left(\bigcap_{n=1}^{\infty} \{X \le x_n\}\right) \\ &= \mathbb{P}(X \le x) \\ &= F(x).\end{aligned}$$

Therefore F is right continuous. □

1.1.30 Example If

$$X \sim \mathrm{Ber}(p),$$

then

$$F(x) = \mathbb{P}(X \le x)$$

is given by

$$F(x) = \begin{cases} 0, & x < 0, \\ 1-p, & 0 \le x < 1, \\ 1, & x \ge 1. \end{cases}$$

1.1.31 Remark For a continuous random variable X with density function f,

$$\mathbb{P}(a < X < b) = F(b) - F(a) = \int_a^b f(x)\,dx.$$

1.1.32 Example A random variable X is uniformly distributed on $[a, b]$, written

$$X \sim \mathrm{Unif}[a, b],$$

if its density function is

$$f(x) = \begin{cases} \dfrac{1}{b-a}, & x \in [a, b], \\ 0, & \text{otherwise.} \end{cases}$$

1.1.33 Definition Let $\lambda > 0$. A random variable X has the **exponential distribution** with parameter λ, written

$$X \sim \mathrm{Exp}(\lambda),$$

if its density function is

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

1.1.34 Proposition If

$$X \sim \text{Exp}(\lambda),$$

then

$$\mathbb{P}(X > t) = e^{-\lambda t}, \qquad t \geq 0.$$

Proof

$$\begin{aligned} \mathbb{P}(X > t) &= \int_t^\infty \lambda e^{-\lambda x}\, dx \\ &= \left[-e^{-\lambda x}\right]_t^\infty \\ &= e^{-\lambda t}. \end{aligned}$$

□

1.1.35 Proposition The exponential distribution satisfies the memoryless property:

$$\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t).$$

Proof

$$\begin{aligned} \mathbb{P}(X > s + t \mid X > s) &= \frac{\mathbb{P}(X > s + t)}{\mathbb{P}(X > s)} \\ &= \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} \\ &= e^{-\lambda t} \\ &= \mathbb{P}(X > t). \end{aligned}$$

□

1.1.36 Exercise Show that the exponential distribution is the only continuous distribution satisfying the memoryless property.

1.1.37 Definition A random variable X has the normal distribution with pa-

rameters $\mu \in \mathbb{R}$ and $\sigma > 0$, written

$$X \sim N(\mu, \sigma^2),$$

if its density function is

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad x \in \mathbb{R}.$$

1.1.38 Remark The constant in the density is chosen so that

$$\int_{\mathbb{R}} f(x)\, dx = 1.$$

It remains to show that

$$\int_{\mathbb{R}} e^{-x^2/2}\, dx = \sqrt{2\pi}.$$

Proof Let

$$I = \int_{\mathbb{R}} e^{-x^2/2}\, dx.$$

Then

$$I^2 = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)/2}\, dxdy.$$

Using polar coordinates,

$$x = r\cos\theta, \qquad y = r\sin\theta,$$

we obtain

$$\begin{aligned} I^2 &= \int_0^{2\pi} \int_0^{\infty} e^{-r^2/2} r\, dr d\theta \\ &= \int_0^{2\pi} \int_0^{\infty} e^{-u}\, du\, d\theta \qquad \left(u = \frac{r^2}{2}\right) \\ &= 2\pi. \end{aligned}$$

Hence

$$I = \sqrt{2\pi}.$$

□

1.1.39 Definition Let X be a random variable.
If X is discrete and

$$\mathbb{P}(X = x_k) = p_k,$$

with

$$\sum_k p_k |x_k| < +\infty,$$

then the expectation of X is defined by

$$\mathbb{E}X := \sum_k p_k x_k.$$

1.1.40 Definition If X has density function f and

$$\int_{\mathbb{R}} |x| f(x)\, dx < +\infty,$$

then

$$\mathbb{E}X := \int_{\mathbb{R}} x f(x)\, dx.$$

1.1.41 Remark More generally, on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, if

$$\int_\Omega |X|\, d\mathbb{P} < +\infty,$$

then

$$\mathbb{E}X = \int_\Omega X\, d\mathbb{P}.$$

1.1.42 Definition The variance of a random variable X is defined by

$$\mathrm{Var}(X) := \mathbb{E}(X - \mathbb{E}X)^2.$$

Equivalently,

$$\mathrm{Var}(X) = \mathbb{E}X^2 - (\mathbb{E}X)^2.$$

1.1.43 Theorem **(Jensen's inequality)** Let

$$J : \mathbb{R} \to \mathbb{R}$$

be convex. Then

$$\mathbb{E}J(X) \geq J(\mathbb{E}X).$$

1.1.44 Example For $p \geq 1$,

$$J(x) = |x|^p$$

is convex. Hence

$$\mathbb{E}|X|^p \geq |\mathbb{E}X|^p.$$

1.1.45 Theorem **(Hölder's inequality)** Let $p, q \in [1, \infty)$ satisfy

$$\frac{1}{p} + \frac{1}{q} = 1.$$

Then

$$|\mathbb{E}(XY)| \leq (\mathbb{E}|X|^p)^{1/p} (\mathbb{E}|Y|^q)^{1/q}.$$